

QUANTITATIVE EQUITY RESEARCH

A-Share Multi-Factor Research System V2

Point-in-Time Universe, Neutralized Factors, Walk-Forward Modelling and Event-Driven Backtesting

Research universe: CSI 300 historical constituents

Report generated: 2026-09-10

Framework: Python · SQLite · pandas · matplotlib

1. Executive Summary

This report evaluates an end-to-end A-share multi-factor research framework designed to reduce survivorship bias, look-ahead bias and unrealistic execution assumptions.

HISTORICAL STOCKS

520

UNIVERSE SNAPSHOTS

N/A

FINAL PORTFOLIO VALUE

¥1,251,689.57

AFTER-COST RETURN

25.17%

EXCESS VS CSI 300

26.89%

MAXIMUM DRAWDOWN

-24.18%

The portfolio outperformed the CSI 300 over the event-backtest period. However, the Sharpe ratio of 0.263 indicates that the strategy still carries meaningful volatility and should not be interpreted as a low-risk investment. Explicit costs consumed 4.29% of initial capital.

2. Research Architecture

The research process separates data, signals, portfolio decisions and execution so that each stage can be independently audited.

STEP 01

Point-in-Time Data

Historical CSI 300 membership, stock master, market data and tradability filters.

STEP 02

Factor Research

Raw factors, winsorization, standardization, industry and size neutralization.

STEP 03

Walk-Forward Model

Rolling historical IC estimates with no future information used in model weights.

STEP 04

Event Execution

Next-open trading, board lots, liquidity constraints, fees, tax and slippage.

3. Data and Universe Quality

The investment universe is reconstructed from historical index snapshots. Delisted securities and former index constituents remain in the database to reduce survivorship bias.

Quality Check	Status	Issues	Description
Minimum stock universe size	PASS	0	At least 250 stocks should be available.
Duplicate stock-date observations	PASS	0	Each stock-date pair must be unique.
Missing identifiers	PASS	0	Date and stock identifiers cannot be missing.
Missing OHLC on trading days	PASS	0	Tradable observations require complete OHLC prices.
Invalid OHLC relationships	PASS	0	High must be the highest and low the lowest price.
Non-positive prices	PASS	0	Trading-day prices must be greater than zero.
Negative volume or turnover	PASS	0	Volume, amount and turnover cannot be negative.
Invalid trading-status flags	PASS	0	Trading status must equal zero or one.
Invalid ST flags	PASS	0	ST status must equal zero or one.
Extreme daily returns above 25%	WARNING	76	May include IPO observations or data anomalies.
Missing valuation indicators	PASS	0	Missing values can occur for loss-making companies.
Stocks with stale ending dates	WARNING	11	The last observation is over 10 days behind.
Stocks with less than one year of data	WARNING	1	Usually caused by recent IPO listings.

Factor Data Coverage

Factor	Category	Role	Coverage	First Valid Snapshot
20-Day Momentum	Momentum	alpha	100.00%	2019-07-31
60-Day Momentum	Momentum	alpha	100.00%	2019-07-31
120-Day Momentum	Momentum	alpha	100.00%	2019-07-31
12-1 Momentum	Momentum	alpha	92.52%	2020-01-31
5-Day Reversal	Reversal	alpha	100.00%	2019-07-31
20-Day Volatility	Risk	alpha	100.00%	2019-07-31
60-Day Downside Volatility	Risk	alpha	100.00%	2019-07-31
60-Day Drawdown	Risk	alpha	100.00%	2019-07-31
60-Day MA Trend	Trend	alpha	100.00%	2019-07-31
20-Day Amount Liquidity	Liquidity	alpha	100.00%	2019-07-31
20-Day Turnover	Liquidity	alpha	100.00%	2019-07-31
Turnover Stability	Liquidity	alpha	100.00%	2019-07-31
Amihud Liquidity	Liquidity	alpha	100.00%	2019-07-31
Earnings Yield Style	Value	alpha	94.53%	2019-07-31
Book-to-Price Style	Value	alpha	100.00%	2019-07-31
Sales-to-Price Style	Value	alpha	99.95%	2019-07-31
Cash-Flow Yield Style	Value	alpha	60.19%	2019-07-31
Log Float Market Capitalization	Size	neutralizer	100.00%	2019-07-31

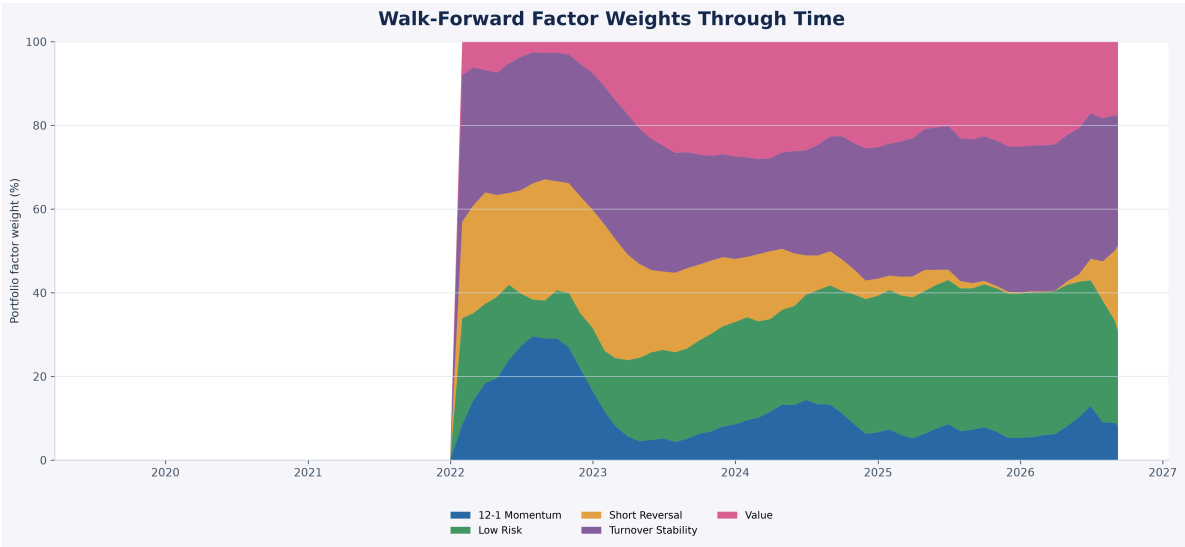
4. Factor Evaluation

Factors are evaluated using cross-sectional Spearman information coefficients and forward returns. The table shows the strongest one-month neutralized signals.

Factor	Variant	Periods	Mean IC	IC IR	Positive IC Rate	t-Statistic
Turnover Stability	neutralized	86	0.029	0.327	66.28%	3.037
60-Day Downside Volatility	neutralized	86	0.027	0.203	61.63%	1.880
20-Day Volatility	neutralized	86	0.026	0.224	60.47%	2.078
12-1 Momentum	neutralized	80	0.022	0.153	63.75%	1.368
Earnings Yield Style	neutralized	86	0.020	0.169	59.30%	1.565
Book-to-Price Style	neutralized	86	0.014	0.108	48.84%	0.998
5-Day Reversal	neutralized	86	0.014	0.133	51.16%	1.237
Sales-to-Price Style	neutralized	86	0.006	0.054	48.84%	0.500
60-Day Drawdown	neutralized	86	0.006	0.048	62.79%	0.442
Cash-Flow Yield Style	neutralized	85	0.005	0.079	51.76%	0.727

Walk-Forward Factor Allocation

Only information available before each rebalance is used to estimate factor weights.



5. Walk-Forward Factor Model

Model weights are estimated from historical IC observations available before each rebalance. Weight smoothing and component limits reduce instability and factor concentration.

Latest Model Weights — 2026-09-08

Component	Training Periods	Weighted Mean IC	Positive IC Rate	Target Weight	Smooth
Turnover Stability	36	0.027	66.67%	27.69%	30.94%
Low Risk	36	0.027	66.67%	17.97%	22.49%
12-1 Momentum	36	0.010	55.56%	6.76%	8.26%
Value	36	0.022	55.56%	18.15%	17.74%
Short Reversal	36	0.032	47.22%	29.44%	20.58%

- Rolling training window: up to 36 months.
- Minimum training history: 24 months.
- Recent observations receive greater weight.
- Single-component weight is capped.
- Reported look-ahead violations: zero.

6. Portfolio Construction

The model selects 20 approximately equal-weighted stocks. Entry and exit rank buffers reduce unnecessary turnover, while industry limits reduce concentration.

Latest Target Portfolio — 2026-09-08

Position	Symbol	Stock	Industry	Model Rank	Score	Status	Weight
1	601360.SS	三六零	I64互联网和相关服务	1	2.592	NEW_ENTRY	5.00%
2	002236.SZ	大华股份	C39计算机、通信和其他电子设备制造业	2	2.572	RETAINED	5.00%
3	002415.SZ	海康威视	C39计算机、通信和其他电子设备制造业	3	2.561	RETAINED	5.00%
4	688009.SS	中国通号	C37铁路、船舶、航空航天和其他运输设备制造业	4	2.176	RETAINED	5.00%
5	600406.SS	国电南瑞	I65软件和信息技术服务业	6	2.034	RETAINED	5.00%
6	002558.SZ	巨人网络	I64互联网和相关服务	7	1.982	NEW_ENTRY	5.00%
7	000157.SZ	中联重科	C35专用设备制造业	9	1.884	RETAINED	5.00%
8	000538.SZ	云南白药	C27医药制造业	11	1.786	RETAINED	5.00%
9	601766.SS	中国中车	C37铁路、船舶、航空航天和其他运输设备制造业	13	1.567	NEW_ENTRY	5.00%
10	600741.SS	华域汽车	C36汽车制造业	14	1.507	RETAINED	5.00%
11	300450.SZ	先导智能	C35专用设备制造业	15	1.490	NEW_ENTRY	5.00%

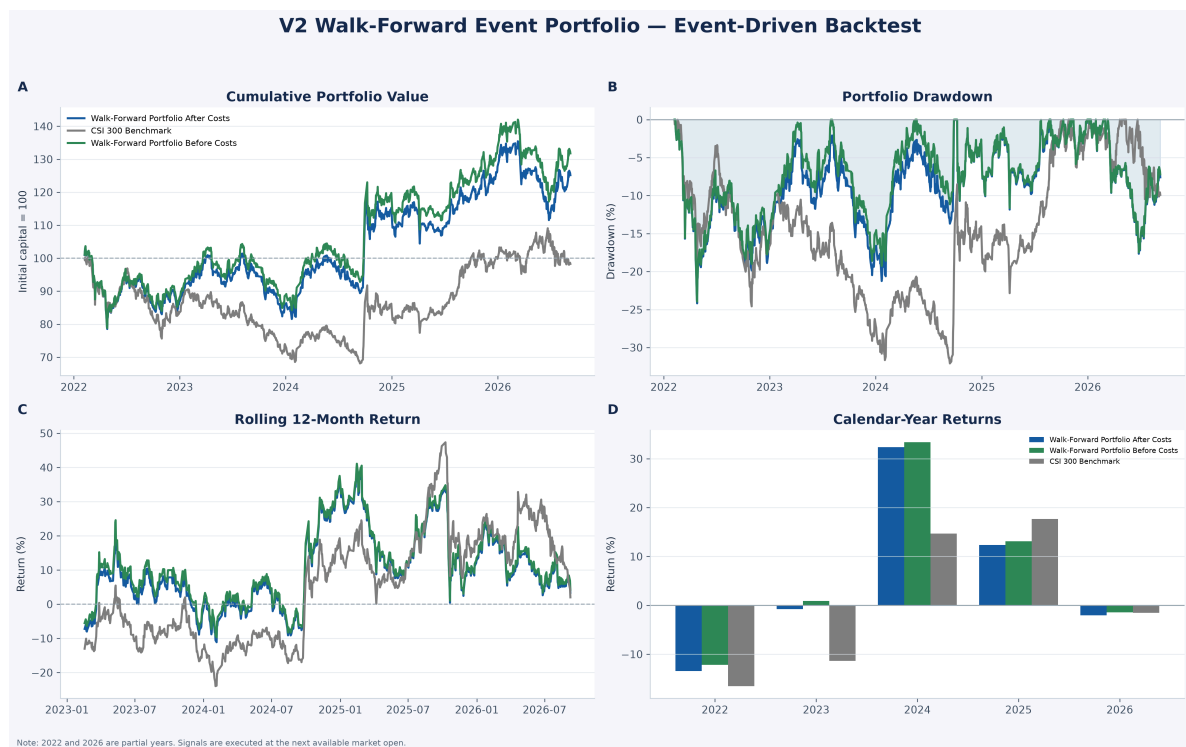
Position	Symbol	Stock	Industry	Model Rank	Score	Status	Weight
12	300628.SZ	亿联网络	C39计算机、通信和其他电子设备制造业	16	1.483	RETAINED	5.00%
13	000999.SZ	华润三九	C27医药制造业	17	1.450	NEW_ENTRY	5.00%
14	600600.SS	青岛啤酒	C15酒、饮料和精制茶制造业	18	1.429	NEW_ENTRY	5.00%
15	000333.SZ	美的集团	C38电气机械和器材制造业	21	1.362	RETAINED	5.00%
16	000651.SZ	格力电器	C38电气机械和器材制造业	22	1.355	RETAINED	5.00%
17	600585.SS	海螺水泥	C30非金属矿物制品业	23	1.323	RETAINED	5.00%
18	600845.SS	宝信软件	I65软件和信息技术服务业	30	1.123	RETAINED	5.00%
19	600570.SS	恒生电子	I65软件和信息技术服务业	36	1.049	RETAINED	5.00%
20	600690.SS	海尔智家	C38电气机械和器材制造业	38	1.035	RETAINED	5.00%

7. Event-Driven Backtest

Signals are formed after the snapshot close and executed at the next available open. Sells are processed before buys, and trades must respect available cash and 100-share board lots.

Portfolio Performance Dashboard

Cumulative value, drawdown, rolling return and calendar-year performance.



Performance Summary

Method	Final Value	Total Return	Annual Return	Annual Volatility	Sharpe	Sortino
Walk-Forward Portfolio After Costs	¥1,251,689.57	25.17%	5.20%	17.74%	0.263	0.374
Walk-Forward Portfolio Before Costs	¥1,317,757.99	31.78%	6.43%	17.74%	0.328	0.469

Method	Final Value	Total Return	Annual Return	Annual Volatility	Sharpe	Sortino
CSI 300 Benchmark	¥982,788.04	-1.72%	-0.39%	18.14%	-0.040	-0.058

Calendar-Year Returns

Year	Portfolio After Costs	Portfolio Before Costs	CSI 300
2022.0	-13.43%	-12.20%	-16.53%
2023.0	-0.77%	0.92%	-11.38%
2024.0	32.38%	33.37%	14.68%
2025.0	12.38%	13.12%	17.66%
2026.0	-2.06%	-1.42%	-1.54%

8. Risk and Relative Performance

Performance and Risk Comparison

Comparison of returns, volatility, Sharpe ratio and maximum drawdown.



Relative Performance

Comparison	Cumulative Excess Return	Tracking Error	Information Ratio	Beta	Annual Alpha
Walk-Forward Portfolio After Costs vs CSI 300	26.89%	10.65%	0.506	0.806	5.25%

Maximum Drawdown Details

Method	Maximum Drawdown	Peak	Trough	Recovery	Peak-to-Trough Days	Recovery Days
Walk-Forward	-24.18%	2022-02-10	2022-04-26	2024-09-30	51	591

Method	Maximum Drawdown	Peak	Trough	Recovery	Peak-to-Trough Days	Recovery Days
Portfolio After Costs						
Walk-Forward Portfolio Before Costs	-23.95%	2022-02-10	2022-04-26	2023-07-31	51	307
CSI 300 Benchmark	-32.09%	2022-02-09	2024-02-12	2025-10-02	634	254

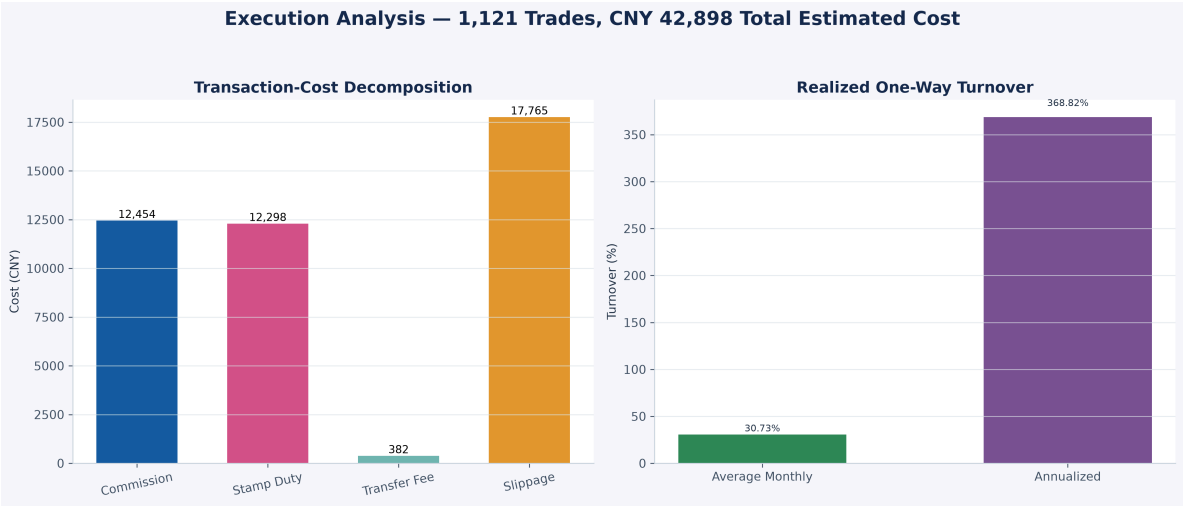


9. Turnover and Trading Costs

The realistic simulation includes commission, minimum commission, stamp duty, transfer fees and bidirectional slippage.

Turnover and Execution Costs

Transaction-cost decomposition and realized portfolio turnover.



Executed Trades	Rebalances	Average Monthly Turnover	Annualized Turnover	Commission
1,121	57	30.73%	368.82%	¥12,454.08

10. Key Conclusions

- Point-in-time membership and inactive-stock retention materially improve research integrity.
- Turnover stability, low-risk factors, value and long-horizon momentum provide the principal model inputs.
- The after-cost portfolio achieved 25.17%, compared with -1.72% for the CSI 300.
- Transaction costs are economically meaningful and reduce the advantage of frequent rebalancing.
- The strategy demonstrates positive relative performance, but its absolute Sharpe ratio and drawdown still require improvement.

11. Limitations

- Historical results do not guarantee future performance.
- The model uses end-of-day data and does not model intraday order-book liquidity.
- Slippage and market impact are simplified and may be higher for large institutional portfolios.
- Corporate actions, unavailable data and historical index reconstruction may contain vendor-specific limitations.
- Factor selection is based on a limited historical sample and may be exposed to model risk.

This project is provided solely for education and quantitative research. It is not investment advice or a recommendation to buy or sell securities.

12. Reproducibility

Core V2 research modules can be rerun from the project root using the following commands.

```
python -m src.data_quality
python -m src.anomaly_review
python -m src.tradability
python -m src.factors_v2
python -m src.factor_processing_v2
python -m src.factor_evaluation_v2
python -m src.walk_forward_model_v2
python -m src.portfolio_construction_v2
python -m src.event_backtester_v2
python -m src.performance_diagnostics_v2
python -m src.visualization_v2
python -m src.report_generator_v2
```